Statistical Analysis of Airline Arrival Delays

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Dataset Description and Apply Goals

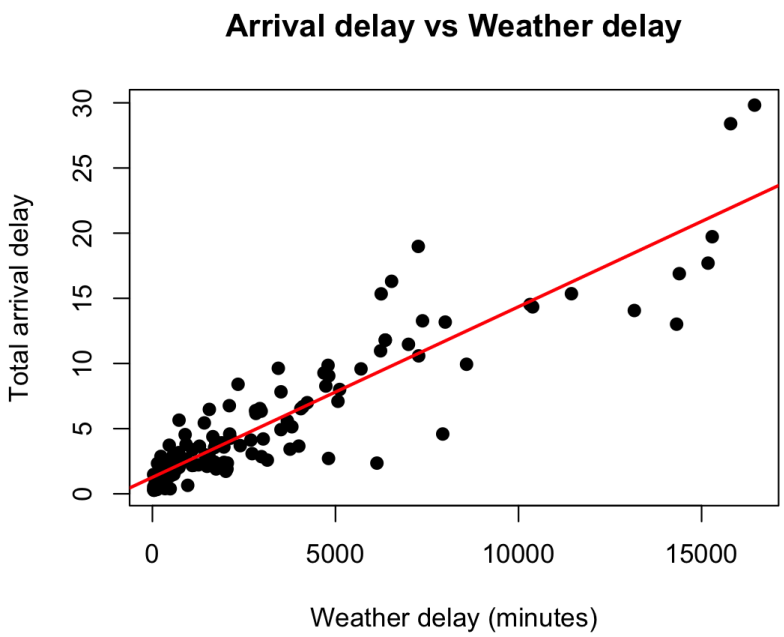
Dataset: Airline On-Time Statistics (BTS)

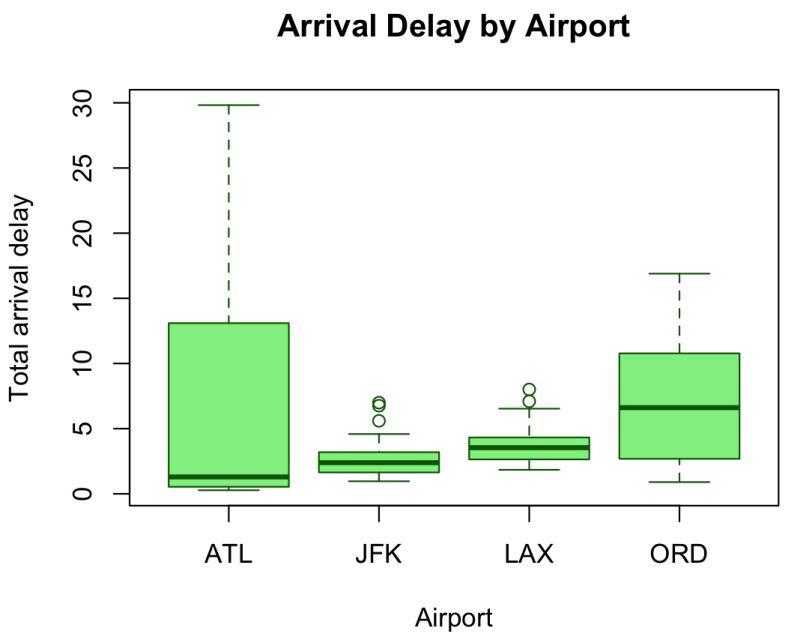
- Observations aggregated by airport–carrier–month
- Sample restricted to:
  - Year: 2019
  - Airports: JFK, LAX, ORD, ATL
  - Carriers: AA, DL, UA

Goal: Identify how different operational factors contribute to airline arrival delays.

Exploratory Data Analysis (EDA)

- A clear positive relationship emerges between weather delay and total arrival delay (scaled).
- Arrival delays also vary across airports, suggesting different operational conditions.

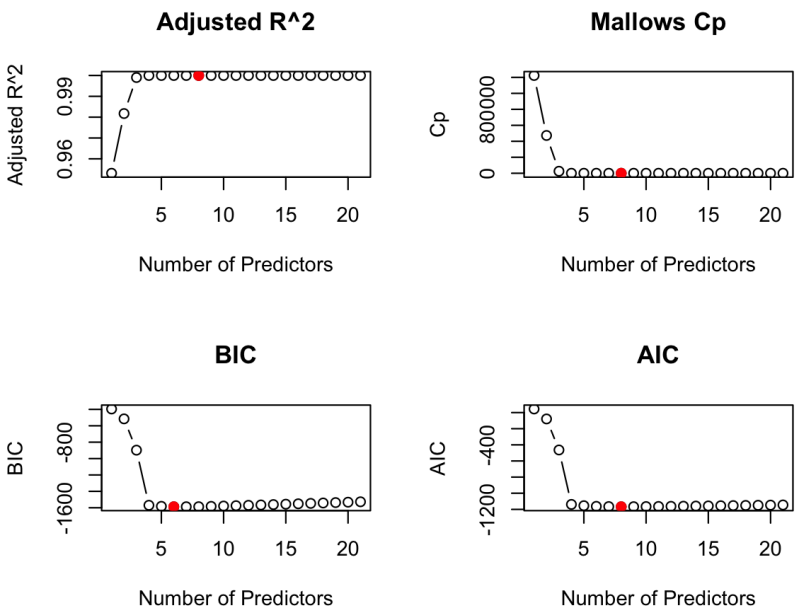


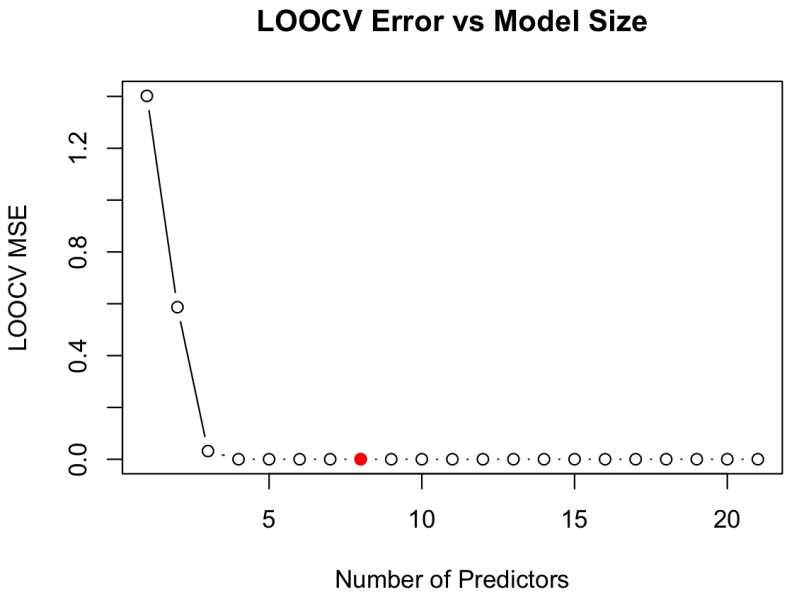


Model Selection and Main Results

Variable selection

- Best subset selection was used to explore different model sizes
- Several criteria were considered: Adjusted R^2, Mallows C_p, BIC and AIC
- Predictive performance was also evaluated using LOOCV

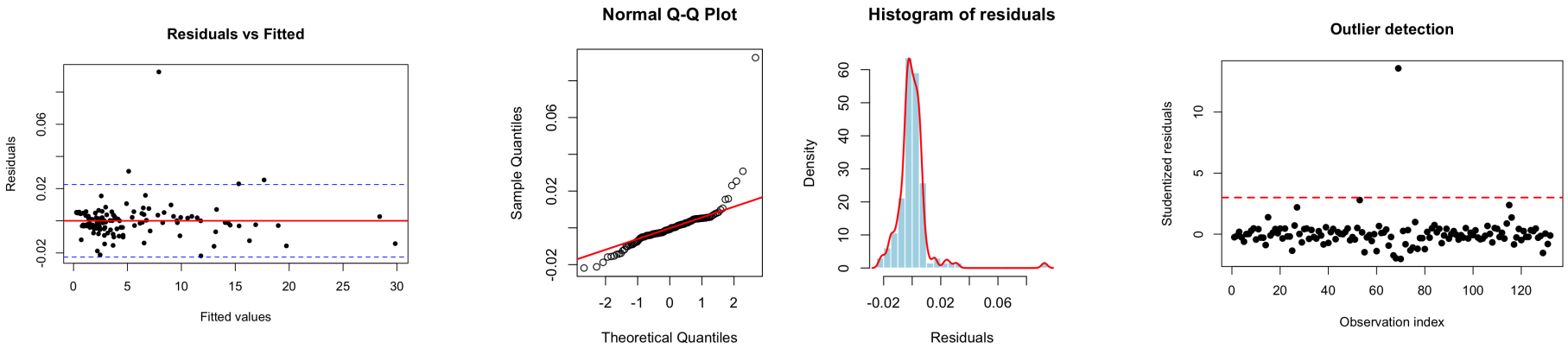




Model Diagnostics

Regression assumptions

- Residual plots show no strong departure from linearity or homoscedasticity
- Residuals appear approximately normally distributed
- Only one observation shows a large studentized residual



Main Results and Conclusions

Objective

- To investigate how different operational delay components explain airline arrival delays.

Main findings

- Weather, carrier, NAS and late aircraft delays are all positive and highly significant predictors of arrival delays.
- Flights departing from LAX show higher delays compared with the baseline airport.
- Delta (DL) and United (UA) exhibit slightly lower delays relative to the reference carrier.
- June is associated with higher average delays.

Limitations and future work

- The analysis is based on aggregated airport–carrier–month data rather than individual flights.
- Future work could extend the analysis using more flexible modelling approaches and additional data to better capture the complexity of delay dynamics.